

The major result in Huang (2008) is that she is able to compute the optimal allocation with tax effects, which industry models rarely do (or if they do, rarely do correctly). Moreover, Huang does it only using a taxable account. Huang's insight is that you can convert any tax-deferred return into a taxable return. A holding of an asset in a tax-deferred account is equivalent to a levered holding of the same asset in a taxable account. We've already seen this in equation (12.6): \$1 contributed pre-tax to a pension fund is worth $1/(1 - 0.396) = \$1.66$ in a taxable account. The taxable portfolio leveres up the tax-deferred portfolio to generate the same return. Anything that the investor can do in a tax-deferred account, the same investor can undo (or re-do) in his taxable account.

Huang's procedure is to compute the equivalent taxable wealth, which is the sum of wealth already in the taxable account, W^T , and then convert all the tax-deferred wealth, W^D , to an equivalent amount in taxable wealth. She redefines total wealth, W , as the sum of

$$w = w^T + Z \times W^D, \quad (12.8)$$

where Z is the appropriate factor that converts the tax-deferred wealth to the taxable equivalent. Most popular industry advice, unfortunately, does not perform optimal allocation calculations on equivalent taxable wealth amounts as per equation (12.8).

Now the standard Merton (1971, 1973) solution applies to total wealth, W , as in chapter 4. The asset allocation and asset location decisions are separable. What makes them so is that there is an appropriate adjustment, Z , which transforms the value of the tax-deferred wealth to taxable wealth and vice versa. The asset allocation problem hence requires an adjustment for the impact of tax deferment.

The key driver in Huang's analysis is the ability to sell short in the taxable account. Small investors will not be able to do this, but large and sophisticated investors certainly can. If investors cannot borrow in the taxable account, then investors may prefer to hold lower taxed assets in the tax-deferred account as this lowers the volatility of the tax benefit. The basic analysis in Huang also misses Constantinides-style tax timing. This can be incorporated. The factor Z in equation (12.8) that converts taxable to tax-deferred wealth can be modified to take into account tax option benefits, where Z now captures *effective* tax rates rather than *statutory* tax rates. In theory, all these effects can be captured in Huang's framework. The downside, however, is that in practice, Huang's Z factor may be difficult to compute.

With the answer to optimal location and optimal asset allocation in hand, we circle back to a question involving muni bonds. If the main benefit of munis is the tax exemption, what about holding munis in a taxable account instead of Treasury or corporate bonds in a tax-deferred account? Dammon, Spatt, and Zhang (2004) examine this question and come to the conclusion that if you